Supplementary Information

Online Resource 1

Machine learning for early prediction of subsequent stage 2–3 acute kidney injury in critically ill adults: hospital-disjoint internal-external validation

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Contents: Supplementary Methods (SM1–SM7), Supplementary Results (SR1–SR7), Tables S1–S18, and Figs. S1–S12.

Supplementary Methods

SM1. Existing sensitivity and heterogeneity analyses

Expanded operating-point scenarios, alternative reference-creatinine definitions, mutually exclusive landmark groups, hospital-level performance and calibration, and outer-fold heterogeneity followed the definitions and held-out evaluation framework described in the main manuscript. These analyses used only information available within the stated temporal windows and did not alter the primary cohort or outcome definition.

SM2. Selection and cohort representativeness

A patient-level comparison universe was constructed to compare the final landmark cohort with candidates lacking a qualifying reference creatinine and candidates excluded after reference-creatinine availability. Groups were mutually exclusive after deterministic selection of one analytic stay per patient. Continuous characteristics were summarized by median and interquartile range; standardized mean differences (SMDs) used observed values, while missingness was reported separately. Hospital reference-creatinine availability was summarized descriptively with denominators; no missingness mechanism or causal interpretation was assigned.

SM3. Sparse-predictor and explicit missingness-signal robustness

Within each outer-training partition, source predictors with at least 50% missingness were removed before preprocessing and the locked nested hospital-disjoint XGBoost workflow was repeated, including training-derived Platt calibration. A separate analysis identified explicit numeric missingness indicators generated by training-fold preprocessing, calculated native TreeSHAP attribution in held-out outer-test patients, and paired indicator contributions with their source-value components. A no-indicator XGBoost model retained median imputation and observed source values but omitted indicator generation. Paired differences used 2,000 hospital-cluster bootstrap replicates.

SM4. Parsimonious renal comparator analysis

Renal comparators used the identical cohort, temporal windows, and hospital-disjoint outer folds. Comparator 0 used reference creatinine alone. Comparator 1, the strongest parsimonious renal-status comparator, used reference creatinine, last creatinine, last-to-reference creatinine ratio, and KDIGO stage 1 status at the landmark. Comparator 2 was specified before performance review as the primary creatinine-trajectory comparator and used reference, last and maximum creatinine, creatinine slope per hour, measurement count, and landmark stage 1 status. Comparator 3 added age, sex, vasopressor treatment, and invasive mechanical ventilation to Comparator 2. Imputation and scaling were fitted only in outer-training hospitals; probabilities were evaluated in held-out hospitals. Incremental discrimination, calibration, operating points, decision curves from 1% to 10%, and matched alert burden used paired hospital-level evaluation and 2,000 hospital-cluster bootstrap replicates where applicable.

SM5. Exploratory subgroup performance

The original fold-specific calibrated held-out XGBoost probabilities were evaluated across sex, prespecified age categories (<50, 50–64, 65–79, and ≥80 years), and recorded race/ethnicity categories without subgroup-specific refitting or recalibration. Discrimination, calibration, prevalence, and operating characteristics at 5%, 7.5%, and 10% thresholds were estimated. Uncertainty used 2,000 hospital-cluster bootstrap samples. Small or event-free groups were retained descriptively and not forced into inferential summaries. This was an exploratory subgroup performance analysis, not a formal fairness assessment.

SM6. Actionable lead-time and outcome-window sensitivity

Events within 6, 12, or 24 hours after the landmark were progressively excluded rather than relabeled as non-events; unchanged held-out predictions were evaluated. Alternative windows assessed horizon dependence without refitting or recalibration.

SM7. Hospital-level heterogeneity

Hospital performance used untouched calibrated outer-test predictions. Publication summaries required at least 20 events and 20 non-events; no hospital-specific fitting, recalibration, optimization, or raw identifier disclosure was performed.

Supplementary Results

SR1. Existing sensitivity and heterogeneity results

Expanded thresholds demonstrated the expected sensitivity-positive predictive value-alert burden trade-off. Alternative reference-creatinine rules produced event rates of approximately 4.94%-5.39% with low label disagreement among evaluable patients. Outer-fold and eligible-hospital results showed broadly stable discrimination with residual variation in precision-recall performance, calibration, and missingness.

SR2. Selection and cohort representativeness

The patient-level comparison universe comprised the final cohort (n=58,491), candidates without qualifying reference creatinine (n=41,614), and other post-reference exclusions (n=26,219). Observable differences were moderate and concentrated in early support, monitoring, and admission context. Across 208 hospitals, median reference-creatinine availability was 62.9% (interquartile range 48.8%-77.8%). These findings bound generalizability to landmark-eligible patients with required renal surveillance and do not exclude unmeasured selection.

SR3. Sparse-predictor and explicit missingness-signal robustness

The high-missingness ablation achieved AUROC 0.8759 and AUPRC 0.3989 versus 0.8758 and 0.3988 for the full model, with essentially unchanged calibration. Across folds, 26 predictors were removed in every fold and six were threshold-dependent. The 153 explicit numeric missingness indicators accounted for 1.89% of total absolute held-out TreeSHAP attribution. The no-indicator model achieved AUROC 0.8771 and AUPRC 0.4013, indicating no material degradation. These tests do not exclude indirect encoding of measurement processes through observed values, timing, treatments, or context.

SR4. Parsimonious renal comparator results

Full XGBoost achieved AUROC 0.8758 and AUPRC 0.3988. Comparator 1 was the strongest parsimonious renal comparator (AUROC 0.8344; AUPRC 0.2883), whereas prespecified Comparator 2 achieved AUROC 0.7839 and AUPRC 0.2507; creatinine slope was unavailable in 71.81% of patients. At 16.78 alerts per 100 admissions, XGBoost identified 2,247 patients with subsequent stage 2–3 AKI versus 2,069 with Comparator 2, a difference of 178 at matched alert burden. XGBoost had greater estimated net benefit at every evaluated threshold from 1% to 10%.

SR5. Exploratory subgroup performance

Across sufficiently large groups, AUROC ranged from 0.830 to 0.898. Patients aged at least 80 years had lower discrimination (AUROC 0.8295) than younger age groups, while calibration remained relatively close to ideal (intercept −0.0207; slope 0.9722). Recorded African American patients had observed risk 8.34% versus mean predicted risk 7.34%, calibration intercept 0.198, and AUROC 0.889, indicating modest underprediction with preserved discrimination. These exploratory findings do not establish fairness, bias absence, or biological differences.

SR6. Actionable lead-time and outcome-window results

Median time to the first qualifying event was 18.3 hours. AUROC/AUPRC were 0.842/0.252 at ≥12 hours and 0.795/0.134 at ≥24 hours; for >12–48 hours they were 0.897/0.382.

SR7. Hospital-level heterogeneity results

Among 51 Tier B hospitals, median AUROC was 0.878 (IQR 0.856–0.899; minimum 0.730); 47/51 were ≥0.80 and none were <0.70. Calibration intercept median was 0.073 (IQR −0.145 to 0.319), with 28/51 absolute intercepts >0.25; slope median was 1.012 (IQR 0.952–1.097).

The strict 29-hospital subset had median/minimum AUROC 0.883/0.823. Removing the 10 highest-AUROC hospitals changed pooled AUROC from 0.876 to 0.870.

Supplementary Tables

Table S1 Expanded operating-point scenarios for calibrated XGBoost. Panel A: performance and alert burden

Threshold	Patients	Events	TP	FP	TN	FN	Sensitivity	Specificity	PPV	NPV	F1	Alerts per 100	True alerts per 100	False alerts per 100	Patients per TP
0.030	58,491	3,032	2,490	12,369	43,090	542	0.821	0.777	0.168	0.988	0.278	25.404	4.257	21.147	5.967
0.050	58,491	3,032	2,247	7,567	47,892	785	0.741	0.864	0.229	0.984	0.350	16.779	3.842	12.937	4.368
0.075	58,491	3,032	2,107	5,690	49,769	925	0.695	0.897	0.270	0.982	0.389	13.330	3.602	9.728	3.701
0.100	58,491	3,032	2,013	4,683	50,776	1,019	0.664	0.916	0.301	0.980	0.414	11.448	3.442	8.006	3.326

Table S1 (continued) Panel B: 95% confidence intervals

Threshold	Sensitivity CI low	Sensitivity CI high	Specificity CI low	Specificity CI high	PPV CI low	PPV CI high	NPV CI low	NPV CI high	F1 CI low	F1 CI high	Alerts per 100, CI low	Alerts per 100, CI high	False alerts per 100, CI low	False alerts per 100, CI high
0.030	0.805	0.837	0.767	0.787	0.158	0.176	0.986	0.989	0.265	0.291	24.363	26.424	20.193	22.102
0.050	0.724	0.759	0.858	0.870	0.219	0.239	0.983	0.985	0.337	0.362	16.089	17.462	12.350	13.505
0.075	0.679	0.711	0.892	0.902	0.258	0.282	0.980	0.983	0.376	0.403	12.776	13.875	9.275	10.191
0.100	0.648	0.680	0.911	0.920	0.287	0.313	0.979	0.982	0.400	0.427	10.953	11.933	7.603	8.386

Values displayed to three decimal places. Thresholds are calibrated probability cut-offs. TP, true positives; FP, false positives; TN, true negatives; FN, false negatives; PPV, positive predictive value; NPV, negative predictive value; CI, confidence interval.

Table S2 Sensitivity analysis of alternative reference-creatinine definitions

Reference-creatinine rule	Primary cohort patients	Reference available	Severe AKI by landmark	Alternative eligible	Alternative events	Alternative event rate	Labels agree	Labels disagree	Changed 0→1	Changed 1→0	Reference coverage	Disagreement rate among alternative eligible
first_icu_0_360	58,491	38,244	142	38,102	1,883	0.049	37,905	197	33	164	0.654	0.005
locked_current	58,491	58,491	0	58,491	3,032	0.052	58,491	0	0	0	1	0
minimum_prelandmark	58,491	58,491	594	57,897	3,048	0.053	57,788	109	109	0	1	0.002
pre_icu_earliest	58,491	33,318	0	33,318	1,795	0.054	33,318	0	0	0	0.570	0

Rules are repository identifiers of the alternative reference-creatinine definitions; locked_current is the primary rule used in the main analysis. Rates and coverages are proportions of the eligible patients under each rule.

Table S3 Mutually exclusive landmark selection groups

Group	Patients	Observed stage 2–3 events	Mean age, years	Median age, years	Female fraction	ICU mortality fraction	Median ICU hours	Mean reference creatinine, mg/dL	Stage 1 fraction	Candidate fraction
included_main	58,491	3,032	63.469	65	0.450	0.042	57.183	1.451	0.050	0.463
no_reference_creatinine	41,614	0	63.964	66	0.470	0.032	37.817	—	0	0.329
no_future_creatinine	18,066	0	61.528	63	0.474	0.001	20.200	1.188	0.022	0.143
inadequate_followup	2,948	0	63.090	65	0.468	0.032	72.417	1.129	0.032	0.023
early_death	2,181	0	69.478	72	0.475	1	30.683	1.837	0.172	0.017
chronic_dialysis_or_esrd	1,635	748	61.846	63	0.428	0.104	50.067	7.044	0.042	0.013
severe_aki_by_12h	1,113	655	61.648	63	0.359	0.181	67.083	4.661	0	0.009
acute_rrt_by_12h	276	49	60.366	61	0.457	0.221	83.850	5.447	0.120	0.002

Groups are repository identifiers of the mutually exclusive landmark selection and exclusion groups; included_main is the final primary cohort. Counts are defined within the landmark candidate set.

Table S4 Hospital-level heterogeneity summary

Measure	Value
All hospitals	198
Hospitals with ≥20 events and ≥20 non-events	51
Median AUROC	0.878
AUROC, IQR lower	0.856
AUROC, IQR upper	0.899
Median AUPRC	0.448
AUPRC, IQR lower	0.389
AUPRC, IQR upper	0.506
Median calibration gap	−0.003
Calibration gap, IQR lower	−0.009
Calibration gap, IQR upper	0.005

Hospital-level estimates use untouched calibrated outer-test predictions; Tier B hospitals are those with at least 20 events and 20 non-events.

Table S5 Selection characteristics and missingness by landmark-selection group. Panel A: characteristics and absolute standardized mean differences

Characteristic	Final primary cohort	Reference creatinine unavailable	Other post-reference exclusions	Missingness by group	Abs SMD final vs no reference	Abs SMD final vs other excluded
Age, years	65.0 (53.0–76.0)	66.0 (54.0–77.0)	64.0 (52.0–76.0)	See Panel B	0.029	0.063
Body mass index, kg/m²	27.6 (23.6–32.9)	27.6 (23.7–32.7)	27.4 (23.3–32.6)	See Panel B	0.005	0.046
Last heart rate by 12 h, beats/min	84.0 (72.0–97.0)	79.0 (68.0–91.0)	81.0 (70.0–94.0)	See Panel B	0.252	0.128
Last systolic blood pressure by 12 h, mmHg	116.0 (103.0–133.0)	119.0 (105.0–136.0)	119.0 (105.0–136.0)	See Panel B	0.120	0.096
Last respiratory rate by 12 h, breaths/min	18.0 (15.0–22.0)	18.0 (15.0–22.0)	18.0 (15.0–22.0)	See Panel B	0.053	0.006
Last blood urea nitrogen by 12 h, mg/dL	19.0 (13.0–32.0)	17.0 (12.0–27.0)	18.0 (12.0–31.0)	See Panel B	0.178	0.033
Any vasopressor infusion by 12 h	8,374 (14.3%)	2,842 (6.8%)	2,489 (9.5%)	0 (0.0%) in each group	0.245	0.149
Invasive mechanical ventilation by 12 h	17,077 (29.2%)	5,730 (13.8%)	4,413 (16.8%)	0 (0.0%) in each group	0.382	0.297
Sex: Female	26,304 (45.0%)	19,576 (47.0%)	12,197 (46.5%)	Missing retained as category	0.042	0.031
Sex: Male	32,179 (55.0%)	22,009 (52.9%)	14,011 (53.4%)	Missing retained as category	0.043	0.032
Sex: Other/missing	8 (0.0%)	29 (0.1%)	11 (0.0%)	Missing retained as category	0.027	0.017
Race/ethnicity: African American	6,394 (10.9%)	3,772 (9.1%)	3,243 (12.4%)	Missing retained as category	0.062	0.045
Race/ethnicity: Asian	1,069 (1.8%)	603 (1.4%)	445 (1.7%)	Missing retained as category	0.030	0.010
Race/ethnicity: Caucasian	44,858 (76.7%)	33,193 (79.8%)	19,635 (74.9%)	Missing retained as category	0.074	0.042
Race/ethnicity: Hispanic	2,214 (3.8%)	1,476 (3.5%)	1,026 (3.9%)	Missing retained as category	0.013	0.007
Race/ethnicity: Missing	505 (0.9%)	679 (1.6%)	334 (1.3%)	Missing retained as category	0.069	0.040
Race/ethnicity: Native American	383 (0.7%)	270 (0.6%)	180 (0.7%)	Missing retained as category	0.001	0.004
Race/ethnicity: Other/Unknown	3,068 (5.2%)	1,621 (3.9%)	1,356 (5.2%)	Missing retained as category	0.065	0.003

Characteristic	Final primary cohort	Reference creatinine unavailable	Other post-reference exclusions	Missingness by group	Abs SMD final vs no reference	Abs SMD final vs other excluded
ICU type: CCU-CTICU	5,028 (8.6%)	3,484 (8.4%)	2,053 (7.8%)	Missing retained as category	0.008	0.028
ICU type: CSICU	2,319 (4.0%)	1,627 (3.9%)	754 (2.9%)	Missing retained as category	0.003	0.060
ICU type: CTICU	2,128 (3.6%)	1,450 (3.5%)	698 (2.7%)	Missing retained as category	0.008	0.056
ICU type: Cardiac ICU	4,048 (6.9%)	2,342 (5.6%)	2,141 (8.2%)	Missing retained as category	0.053	0.047
ICU type: MICU	5,334 (9.1%)	3,150 (7.6%)	2,252 (8.6%)	Missing retained as category	0.056	0.019
ICU type: Med-Surg ICU	31,585 (54.0%)	23,296 (56.0%)	14,508 (55.3%)	Missing retained as category	0.040	0.027
ICU type: Neuro ICU	3,785 (6.5%)	4,167 (10.0%)	2,154 (8.2%)	Missing retained as category	0.129	0.067
ICU type: SICU	4,264 (7.3%)	2,098 (5.0%)	1,659 (6.3%)	Missing retained as category	0.094	0.038
ICU admission source: Acute Care/Floor	1,725 (2.9%)	922 (2.2%)	690 (2.6%)	Missing retained as category	0.046	0.019
ICU admission source: Chest Pain Center	117 (0.2%)	108 (0.3%)	52 (0.2%)	Missing retained as category	0.012	0.000
ICU admission source: Direct Admit	4,412 (7.5%)	2,870 (6.9%)	2,020 (7.7%)	Missing retained as category	0.025	0.006
ICU admission source: Emergency Department	26,841 (45.9%)	23,551 (56.6%)	13,702 (52.3%)	Missing retained as category	0.215	0.128
ICU admission source: Floor	8,482 (14.5%)	2,473 (5.9%)	3,486 (13.3%)	Missing retained as category	0.285	0.035
ICU admission source: ICU	82 (0.1%)	267 (0.6%)	56 (0.2%)	Missing retained as category	0.080	0.017
ICU admission source: ICU to SDU	151 (0.3%)	797 (1.9%)	123 (0.5%)	Missing retained as category	0.160	0.035
ICU admission source: Missing	87 (0.1%)	91 (0.2%)	53 (0.2%)	Missing retained as category	0.016	0.013
ICU admission source: Observation	2 (0.0%)	3 (0.0%)	4 (0.0%)	Missing retained as category	0.005	0.012
ICU admission source: Operating Room	10,552 (18.0%)	5,684 (13.7%)	3,053 (11.6%)	Missing retained as category	0.120	0.181
ICU admission source: Other	3 (0.0%)	2 (0.0%)	2 (0.0%)	Missing retained as category	0.000	0.003
ICU admission source: Other Hospital	1,553 (2.7%)	698 (1.7%)	575 (2.2%)	Missing retained as category	0.067	0.030
ICU admission source: Other ICU	395 (0.7%)	445 (1.1%)	192 (0.7%)	Missing retained as category	0.042	0.007
ICU admission source: PACU	474 (0.8%)	586 (1.4%)	264 (1.0%)	Missing retained as category	0.057	0.021
ICU admission source: Recovery Room	2,256 (3.9%)	2,492 (6.0%)	1,425 (5.4%)	Missing retained as category	0.099	0.075
ICU admission source: StepDown Unit (SDU)	1,359 (2.3%)	625 (1.5%)	522 (2.0%)	Missing retained as category	0.060	0.023
Hospital admission source: Acute Care/Floor	1,230 (2.1%)	432 (1.0%)	505 (1.9%)	Missing retained as category	0.086	0.013
Hospital admission source: Chest Pain Center	108 (0.2%)	94 (0.2%)	51 (0.2%)	Missing retained as category	0.009	0.002
Hospital admission source: Direct Admit	3,731 (6.4%)	2,433 (5.8%)	1,704 (6.5%)	Missing retained as category	0.022	0.005
Hospital admission source: Emergency Department	20,251 (34.6%)	18,903 (45.4%)	10,784 (41.1%)	Missing retained as category	0.222	0.134
Hospital admission source: Floor	6,393 (10.9%)	1,912 (4.6%)	2,670 (10.2%)	Missing retained as category	0.238	0.024
Hospital admission source: ICU	12 (0.0%)	14 (0.0%)	7 (0.0%)	Missing retained as category	0.008	0.004

Characteristic	Final primary cohort	Reference creatinine unavailable	Other post-reference exclusions	Missingness by group	Abs SMD final vs no reference	Abs SMD final vs other excluded
Hospital admission source: ICU to SDU	30 (0.1%)	23 (0.1%)	9 (0.0%)	Missing retained as category	0.002	0.008
Hospital admission source: Missing	15,440 (26.4%)	10,356 (24.9%)	6,312 (24.1%)	Missing retained as category	0.035	0.054
Hospital admission source: Observation	3 (0.0%)	4 (0.0%)	3 (0.0%)	Missing retained as category	0.005	0.007
Hospital admission source: Operating Room	7,114 (12.2%)	4,214 (10.1%)	2,073 (7.9%)	Missing retained as category	0.065	0.142
Hospital admission source: Other	4 (0.0%)	1 (0.0%)	1 (0.0%)	Missing retained as category	0.007	0.004
Hospital admission source: Other Hospital	1,057 (1.8%)	560 (1.3%)	379 (1.4%)	Missing retained as category	0.037	0.029
Hospital admission source: Other ICU	114 (0.2%)	52 (0.1%)	55 (0.2%)	Missing retained as category	0.018	0.003
Hospital admission source: PACU	364 (0.6%)	451 (1.1%)	214 (0.8%)	Missing retained as category	0.050	0.023
Hospital admission source: Recovery Room	1,659 (2.8%)	1,948 (4.7%)	1,095 (4.2%)	Missing retained as category	0.097	0.073
Hospital admission source: Step-Down Unit (SDU)	981 (1.7%)	217 (0.5%)	357 (1.4%)	Missing retained as category	0.111	0.026

Continuous variables are median (IQR); categorical variables are n (%). SMDs are descriptive and based on observed values. Panel B reports missingness separately.

Table S5 (continued) Panel B: group-specific missingness

Characteristic	Group	Observed, n	Missing, n	Missing, %
Age, years	A: final primary cohort	58,491	0	0.0%
Age, years	B: no reference creatinine	41,614	0	0.0%
Age, years	C: other post-reference exclusions	26,219	0	0.0%
Body mass index, kg/m²	A: final primary cohort	57,098	1,393	2.4%
Body mass index, kg/m²	B: no reference creatinine	39,174	2,440	5.9%
Body mass index, kg/m²	C: other post-reference exclusions	25,077	1,142	4.4%
Last heart rate by 12 h, beats/min	A: final primary cohort	57,593	898	1.5%
Last heart rate by 12 h, beats/min	B: no reference creatinine	40,628	986	2.4%
Last heart rate by 12 h, beats/min	C: other post-reference exclusions	25,531	688	2.6%
Last systolic blood pressure by 12 h, mmHg	A: final primary cohort	57,614	877	1.5%
Last systolic blood pressure by 12 h, mmHg	B: no reference creatinine	40,554	1,060	2.5%
Last systolic blood pressure by 12 h, mmHg	C: other post-reference exclusions	25,574	645	2.5%
Last respiratory rate by 12 h, breaths/min	A: final primary cohort	53,306	5,185	8.9%
Last respiratory rate by 12 h, breaths/min	B: no reference creatinine	37,983	3,631	8.7%
Last respiratory rate by 12 h, breaths/min	C: other post-reference exclusions	23,203	3,016	11.5%
Last blood urea nitrogen by 12 h, mg/dL	A: final primary cohort	45,473	13,018	22.3%
Last blood urea nitrogen by 12 h, mg/dL	B: no reference creatinine	18,042	23,572	56.6%
Last blood urea nitrogen by 12 h, mg/dL	C: other post-reference exclusions	21,543	4,676	17.8%

Groups: A, final primary cohort; B, no qualifying reference creatinine; C, other post-reference exclusions.

Table S6 Full XGBoost versus training-fold-specific high-missingness ablation

Metric	Full XGBoost	High-missingness ablation	Difference (ablation − full)	95% CI low	95% CI high
AUROC	0.87580	0.87587	0.00007	−0.00136	0.00149
AUPRC	0.39883	0.39893	0.00009	−0.00505	0.00492
Brier score	0.03780	0.03781	0.00001	−0.00013	0.00015
Calibration intercept	0.00227	0.00595	0.00368	−0.00775	0.01474
Calibration slope	1.00458	1.00730	0.00272	−0.00152	0.00744

Differences are ablation minus full model; 95% confidence intervals use 2,000 paired hospital-cluster bootstrap replicates. No formal equivalence margin was tested.

Table S7 Source predictors excluded at the ≥50% outer-training-fold missingness threshold

Predictor	Clinical domain	Overall missingness	Hospitals completely unavailable	Removed in folds	Removed fold count
x_lab_potassium_delta	Laboratory	63.9%	5	1,2,3,4,5	5
x_lab_sodium_delta	Laboratory	67.7%	5	1,2,3,4,5	5
x_lab_magnesium_first	Laboratory	49.9%	4	4,5	2
x_lab_magnesium_last	Laboratory	49.9%	4	4,5	2
x_lab_magnesium_min	Laboratory	49.9%	4	4,5	2
x_lab_magnesium_max	Laboratory	49.9%	4	4,5	2
x_lab_magnesium_mean	Laboratory	49.9%	4	4,5	2
x_lab_magnesium_n	Laboratory	49.9%	4	4,5	2
x_lab_phosphate_first	Laboratory	67.5%	9	1,2,3,4,5	5
x_lab_phosphate_last	Laboratory	67.5%	9	1,2,3,4,5	5
x_lab_phosphate_min	Laboratory	67.5%	9	1,2,3,4,5	5
x_lab_phosphate_max	Laboratory	67.5%	9	1,2,3,4,5	5
x_lab_phosphate_mean	Laboratory	67.5%	9	1,2,3,4,5	5
x_lab_phosphate_n	Laboratory	67.5%	9	1,2,3,4,5	5
x_lab_albumin_first	Laboratory	62.2%	2	1,2,3,4,5	5
x_lab_albumin_last	Laboratory	62.2%	2	1,2,3,4,5	5
x_lab_albumin_min	Laboratory	62.2%	2	1,2,3,4,5	5
x_lab_albumin_max	Laboratory	62.2%	2	1,2,3,4,5	5
x_lab_albumin_mean	Laboratory	62.2%	2	1,2,3,4,5	5
x_lab_albumin_n	Laboratory	62.2%	2	1,2,3,4,5	5
x_lab_bilirubin_total_first	Laboratory	65.2%	6	1,2,3,4,5	5
x_lab_bilirubin_total_last	Laboratory	65.2%	6	1,2,3,4,5	5
x_lab_bilirubin_total_min	Laboratory	65.2%	6	1,2,3,4,5	5
x_lab_bilirubin_total_max	Laboratory	65.2%	6	1,2,3,4,5	5
x_lab_bilirubin_total_mean	Laboratory	65.2%	6	1,2,3,4,5	5
x_lab_bilirubin_total_n	Laboratory	65.2%	6	1,2,3,4,5	5
x_lab_inr_first	Laboratory	64.7%	8	1,2,3,4,5	5
x_lab_inr_last	Laboratory	64.7%	8	1,2,3,4,5	5
x_lab_inr_min	Laboratory	64.7%	8	1,2,3,4,5	5
x_lab_inr_max	Laboratory	64.7%	8	1,2,3,4,5	5
x_lab_inr_mean	Laboratory	64.7%	8	1,2,3,4,5	5
x_lab_inr_n	Laboratory	64.7%	8	1,2,3,4,5	5

Twenty-six predictors were removed in all folds and six were fold-dependent. Missingness decisions were based only on outer-training hospitals.

Table S8 Leading explicit numeric missingness-indicator attributions in held-out XGBoost predictions

Rank	Source predictor	Mean |SHAP|, indicator	Mean |SHAP|, value	Median fold rank	Indicator share of source |SHAP|	Overall missingness
1	x_lab_creatinine_first	0.01590	0.00940	1	62.9%	21.8%
2	x_lab_inr_first	0.00900	0.00540	3	62.5%	64.7%
3	x_vital_noninvasive_mean_bp_sd	0.00626	0.00954	9	39.6%	6.7%
4	x_lab_chloride_first	0.00316	0.00199	20	61.4%	22.1%
5	x_vital_noninvasive_mean_bp_first	0.00299	0.01938	9	13.4%	5.7%
6	x_vital_noninvasive_diastolic_bp_sd	0.00291	0.01060	7	21.6%	6.7%
7	x_vital_noninvasive_systolic_bp_sd	0.00253	0.04464	7	5.4%	6.7%
8	x_lab_creatinine_last	0.00216	0.31628	25	0.7%	21.8%
9	x_lab_calcium_first	0.00194	0.00245	7	44.2%	24.4%
10	x_lab_phosphate_first	0.00142	0.00120	17	54.3%	67.5%
11	x_lab_albumin_first	0.00132	0.00072	11	64.8%	62.2%
12	x_lab_sodium_first	0.00098	0.00310	20	24.0%	21.2%
13	x_lab_inr_last	0.00091	0.00954	17	8.7%	64.7%
14	x_lab_bun_first	0.00090	0.06201	23	1.4%	22.3%
15	x_lab_chloride_last	0.00083	0.01096	26	7.1%	22.1%

Across 153 indicators, the aggregate indicator share was 1.89% of total absolute held-out native TreeSHAP attribution. The table shows the 15 leading indicators; the complete inventory remains reproducibility material.

Table S9 Full XGBoost versus XGBoost without explicit numeric missingness indicators

Model	AUROC	AUPRC	Brier score	Log loss	Calibration intercept	Calibration slope	Mean predicted risk
Full XGBoost	0.87580	0.39883	0.03780	0.14202	0.00227	1.00458	0.05210
No missingness indicators	0.87713	0.40132	0.03773	0.14164	0.00674	1.00718	0.05215

Both models used held-out hospital-disjoint evaluation and training-derived calibration. Paired uncertainty is reported in the underlying analysis outputs.

Table S10 Performance of full XGBoost and renal Comparator 0–3 models

Model	Predictors	AUROC	AUPRC	Brier score	Calibration intercept	Calibration slope
Full XGBoost	159-predictor primary model	0.8758	0.3988	0.0378	0.0023	1.0046
Comparator 0: reference creatinine	x_reference_creatinine	0.7140	0.2057	0.0467	−0.0023	0.9987
Comparator 1: renal status	x_reference_creatinine, x_lab_creatinine_last, x_creatinine_last_reference_ratio, x_stage1_at_landmark	0.8344	0.2883	0.0431	−0.0053	0.9977
Comparator 2: creatinine trajectory	x_reference_creatinine, x_lab_creatinine_last, x_lab_creatinine_max, x_creatinine_slope_per_hour, x_lab_creatinine_n, x_stage1_at_landmark	0.7839	0.2507	0.0440	−0.0172	0.9938
Comparator 3: renal + minimal clinical	x_reference_creatinine, x_lab_creatinine_last, x_lab_creatinine_max, x_creatinine_slope_per_hour, x_lab_creatinine_n, x_stage1_at_landmark, x_age_years, x_sex, x_tx_any_vasopressor, x_resp_invasive_mechanical_ventilation	0.8245	0.2718	0.0434	−0.0182	0.9925

Comparator 1 was the strongest parsimonious renal-status comparator. Comparator 2 was the prespecified creatinine-trajectory comparator and was not the best-performing simple comparator. Within the eligible cohort, the subgroup with no KDIGO stage 1 at the landmark corresponds to creatinine stage 0 (n=55,542; 2,251 events); XGBoost AUROC/AUPRC were 0.8585/0.3506 versus 0.7411/0.2320 for Comparator 2.

Table S11 Paired incremental performance of full XGBoost versus prespecified Comparator 2

Metric	XGBoost	Comparator 2	Difference (XGBoost − Comparator 2)	95% CI low	95% CI high	Bootstrap replicates
AUROC	0.87580	0.78392	0.09188	0.08211	0.10266	2,000
AUPRC	0.39883	0.25068	0.14815	0.12983	0.16664	2,000
Brier score	0.03780	0.04398	−0.00619	−0.00690	−0.00548	2,000
Log loss	0.14202	0.16886	−0.02684	−0.02916	−0.02445	2,000
Calibration intercept	0.00227	−0.01717	0.01944	−0.09262	0.14230	2,000
Calibration slope	1.00458	0.99376	0.01082	−0.02840	0.05331	2,000

Differences are XGBoost minus Comparator 2. Confidence intervals use 2,000 paired hospital-cluster bootstrap replicates.

Table S12 Fixed-threshold and matched-alert comparisons of full XGBoost and prespecified Comparator 2

Model	Comparison	Threshold	Sensitivity	Specificity	PPV	NPV	F1	Alerts per 100	True positives	False positives
Full XGBoost	same probability threshold	0.0500	0.7411	0.8636	0.2290	0.9839	0.3498	16.7786	2,247	7,567
Comparator 2: creatinine trajectory	same probability threshold	0.0500	0.6672	0.8808	0.2344	0.9798	0.3469	14.7561	2,023	6,608
Full XGBoost	same probability threshold	0.0750	0.6949	0.8974	0.2702	0.9818	0.3891	13.3303	2,107	5,690
Comparator 2: creatinine trajectory	same probability threshold	0.0750	0.6069	0.9229	0.3009	0.9772	0.4023	10.4563	1,840	4,276
Full XGBoost	same probability threshold	0.1000	0.6639	0.9156	0.3006	0.9803	0.4139	11.4479	2,013	4,683
Comparator 2: creatinine trajectory	same probability threshold	0.1000	0.5300	0.9358	0.3108	0.9733	0.3919	8.8390	1,607	3,563
Full XGBoost	matched alert burden	0.0500	0.7411	0.8636	0.2290	0.9839	0.3498	16.7786	2,247	7,567
Comparator 2: creatinine trajectory	matched alert burden	0.0448	0.6824	0.8603	0.2108	0.9802	0.3221	16.7786	2,069	7,745

Matched alert burden used the full-model 5% threshold burden (16.78 alerts per 100). Counts are held-out true positives and false positives; they do not represent prevented outcomes.

Table S13 Exploratory subgroup discrimination and calibration

Domain	Subgroup	n	Events	Prevalence	Mean predicted risk	AUROC	AUPRC	Calibration intercept	Calibration slope
Sex	Female	26,304	1,250	4.75%	4.87%	0.860	0.357	−0.082	0.977
Sex	Male	32,179	1,782	5.54%	5.49%	0.886	0.428	0.058	1.024
Sex	Other/missing	8	0	0.00%	4.03%	—	—	—	—
Age	<50	11,217	541	4.82%	4.92%	0.898	0.413	0.009	1.020
Age	50–64	16,826	945	5.62%	5.69%	0.887	0.438	0.014	1.017
Age	65–79	19,472	1,047	5.38%	5.42%	0.874	0.396	−0.005	1.003
Age	≥80	10,976	499	4.55%	4.39%	0.830	0.322	−0.021	0.972
Recorded race/ethnicity	Caucasian	44,858	2,084	4.65%	4.86%	0.871	0.366	−0.058	1.001
Recorded race/ethnicity	African American	6,394	533	8.34%	7.34%	0.889	0.491	0.198	1.002
Recorded race/ethnicity	Other/Unknown	3,068	180	5.87%	5.35%	0.872	0.460	0.084	0.975
Recorded race/ethnicity	Hispanic	2,214	114	5.15%	5.44%	0.884	0.421	−0.041	1.018
Recorded race/ethnicity	Asian	1,069	63	5.89%	5.75%	0.891	0.509	0.143	1.058
Recorded race/ethnicity	Native American	383	27	7.05%	7.26%	0.840	0.437	−0.259	0.871
Recorded race/ethnicity	Missing	505	31	6.14%	5.12%	0.835	0.469	0.152	0.952

Panelled domains include sex, prespecified age groups, and recorded race/ethnicity. Original held-out probabilities were used without subgroup refitting or recalibration. The other/missing sex category (n=8; zero events) is descriptive only. Recorded categories must not be interpreted biologically.

Table S14 Exploratory subgroup operating-point performance at 5%, 7.5%, and 10%

Domain	Subgroup	Threshold	n	Events	Sensitivity	Specificity	PPV	Alerts per 100
Sex	Female	5.0%	26,304	1,250	69.8%	86.5%	20.5%	16.16
Sex	Female	7.5%	26,304	1,250	64.6%	90.1%	24.6%	12.49
Sex	Female	10.0%	26,304	1,250	61.6%	92.0%	27.8%	10.53
Sex	Male	5.0%	32,179	1,782	77.2%	86.2%	24.7%	17.28
Sex	Male	7.5%	32,179	1,782	72.9%	89.4%	28.8%	14.02
Sex	Male	10.0%	32,179	1,782	69.8%	91.2%	31.7%	12.20
Sex	Other/missing	5.0%	8	0	—	—	—	12.50
Sex	Other/missing	7.5%	8	0	—	—	—	12.50
Sex	Other/missing	10.0%	8	0	—	—	—	12.50
Age	<50	5.0%	11,217	541	76.9%	88.2%	24.9%	14.91
Age	<50	7.5%	11,217	541	72.6%	91.4%	30.0%	11.69
Age	<50	10.0%	11,217	541	68.9%	93.1%	33.5%	9.91
Age	50–64	5.0%	16,826	945	77.5%	85.7%	24.4%	17.85
Age	50–64	7.5%	16,826	945	72.5%	89.2%	28.5%	14.28
Age	50–64	10.0%	16,826	945	70.6%	91.1%	32.2%	12.33
Age	65–79	5.0%	19,472	1,047	74.1%	85.6%	22.6%	17.61
Age	65–79	7.5%	19,472	1,047	69.1%	89.0%	26.4%	14.08
Age	65–79	10.0%	19,472	1,047	65.5%	90.9%	29.0%	12.15
Age	≥80	5.0%	10,976	499	64.7%	86.8%	18.9%	15.57
Age	≥80	7.5%	10,976	499	61.3%	90.1%	22.8%	12.23
Age	≥80	10.0%	10,976	499	57.5%	91.8%	25.1%	10.42
Recorded race/ethnicity	Caucasian	5.0%	44,858	2,084	71.9%	86.9%	21.1%	15.82

Domain	Subgroup	Threshold	n	Events	Sensitivity	Specificity	PPV	Alerts per 100
Recorded race/ethnicity	Caucasian	7.5%	44,858	2,084	67.0%	90.3%	25.1%	12.38
Recorded race/ethnicity	Caucasian	10.0%	44,858	2,084	64.0%	92.0%	28.1%	10.59
Recorded race/ethnicity	African American	5.0%	6,394	533	81.1%	82.3%	29.4%	22.99
Recorded race/ethnicity	African American	7.5%	6,394	533	77.5%	86.0%	33.5%	19.30
Recorded race/ethnicity	African American	10.0%	6,394	533	73.9%	88.5%	36.9%	16.70
Recorded race/ethnicity	Other/Unknown	5.0%	3,068	180	74.4%	87.2%	26.6%	16.43
Recorded race/ethnicity	Other/Unknown	7.5%	3,068	180	68.9%	90.5%	31.2%	12.97
Recorded race/ethnicity	Other/Unknown	10.0%	3,068	180	65.0%	92.0%	33.7%	11.31
Recorded race/ethnicity	Hispanic	5.0%	2,214	114	78.9%	85.7%	23.1%	17.62
Recorded race/ethnicity	Hispanic	7.5%	2,214	114	72.8%	88.8%	26.0%	14.41
Recorded race/ethnicity	Hispanic	10.0%	2,214	114	71.1%	90.8%	29.5%	12.42
Recorded race/ethnicity	Asian	5.0%	1,069	63	79.4%	85.7%	25.8%	18.15
Recorded race/ethnicity	Asian	7.5%	1,069	63	76.2%	88.8%	29.8%	15.06
Recorded race/ethnicity	Asian	10.0%	1,069	63	73.0%	90.5%	32.4%	13.28
Recorded race/ethnicity	Native American	5.0%	383	27	74.1%	82.3%	24.1%	21.67
Recorded race/ethnicity	Native American	7.5%	383	27	74.1%	86.5%	29.4%	17.75
Recorded race/ethnicity	Native American	10.0%	383	27	74.1%	87.9%	31.7%	16.45
Recorded race/ethnicity	Missing	5.0%	505	31	71.0%	88.2%	28.2%	15.45
Recorded race/ethnicity	Missing	7.5%	505	31	71.0%	91.6%	35.5%	12.28
Recorded race/ethnicity	Missing	10.0%	505	31	67.7%	93.5%	40.4%	10.30

Metrics are descriptive held-out results. Differences may reflect prevalence, case mix, hospital composition, measurement, and coding practices; they do not establish fairness or causality. Sensitivity, specificity, and PPV are undefined for the event-free other/missing sex category.

Table S15 Actionable lead-time sensitivity

Scenario	n	Events	Prevalence	AUROC	AUPRC	Slope
L0	58491	3032	5.18%	0.876	0.399	1.005
L6	57238	2375	4.15%	0.858	0.303	0.938
L12	53252	1980	3.72%	0.842	0.252	0.885
L24	43215	1106	2.56%	0.795	0.134	0.736

Scenarios L6, L12, and L24 exclude events occurring within 6, 12, and 24 hours after the landmark; L0 is the primary analysis. Events were excluded, not relabeled; unchanged predictions were used.

Table S16 Alternative outcome-window sensitivity

Window	n	Events	Prevalence	AUROC	AUPRC	Slope
W72	58491	3032	5.18%	0.876	0.399	1.005
W48	58491	2429	4.15%	0.897	0.382	1.072
W24to72	53252	1980	3.72%	0.842	0.252	0.885
W24to48	53252	1377	2.59%	0.868	0.217	0.951

W72 is primary; other rows are descriptive horizon sensitivities.

Table S17 Held-out hospital performance distributions

Tier	Metric	Hospitals	Min	P10	Median	Q1	Q3
Tier B (≥20 events and ≥20 non-events)	AUROC	51	0.730	0.827	0.878	0.856	0.899
Tier B (≥20 events and ≥20 non-events)	AUPRC	51	0.156	0.327	0.448	0.389	0.506
Tier B (≥20 events and ≥20 non-events)	Calibration intercept	51	−1.198	−0.348	0.073	−0.145	0.319
Tier B (≥20 events and ≥20 non-events)	Calibration slope	51	0.616	0.911	1.012	0.952	1.097
Strict (≥30 events and ≥30 non-events)	AUROC	29	0.823	0.850	0.883	0.863	0.903
Strict (≥30 events and ≥30 non-events)	AUPRC	29	0.237	0.376	0.478	0.400	0.512
Strict (≥30 events and ≥30 non-events)	Calibration intercept	29	−0.642	−0.182	0.169	−0.065	0.370
Strict (≥30 events and ≥30 non-events)	Calibration slope	29	0.792	0.939	1.026	0.961	1.106

Tier B requires ≥20 events and ≥20 non-events; strict requires ≥30. P10, 10th percentile; Q1 and Q3, first and third quartiles.

Table S18 Lower-tail hospital audit without identifiers

Audit	Rank	n	Events	Prevalence	AUROC	AUPRC
lowest AUROC	1	393	26	0.066	0.730	0.448
lowest AUROC	2	513	20	0.039	0.760	0.156
lowest AUROC	3	262	20	0.076	0.792	0.507
lowest AUROC	4	602	27	0.045	0.799	0.393
lowest AUROC	5	508	32	0.063	0.823	0.411
lowest AUPRC	1	513	20	0.039	0.760	0.156
lowest AUPRC	2	1269	62	0.049	0.827	0.237
lowest AUPRC	3	672	25	0.037	0.913	0.257
lowest AUPRC	4	697	28	0.040	0.870	0.291
lowest AUPRC	5	1772	99	0.056	0.863	0.322
largest absolute intercept	1	393	26	0.066	0.730	0.448
largest absolute intercept	2	513	20	0.039	0.760	0.156
largest absolute intercept	3	228	25	0.110	0.917	0.676
largest absolute intercept	4	1269	62	0.049	0.827	0.237
largest absolute intercept	5	673	32	0.048	0.905	0.610

Raw hospital identifiers are not included.

Supplementary Figures

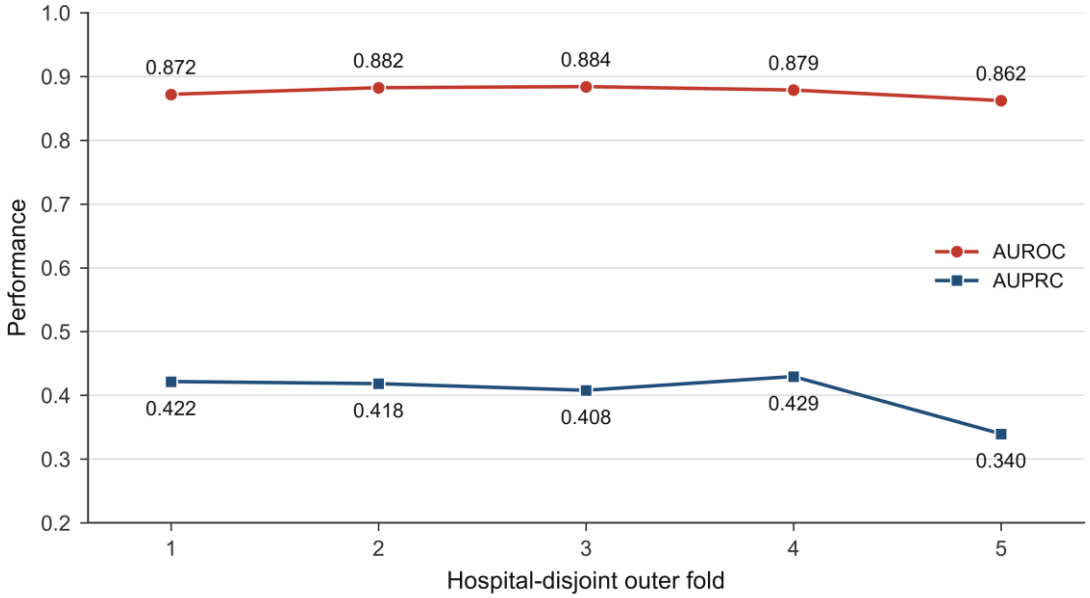


Fig. S1 Hospital-disjoint outer-fold discrimination. Fold-specific AUROC and AUPRC demonstrate performance variation across the five held-out hospital groups

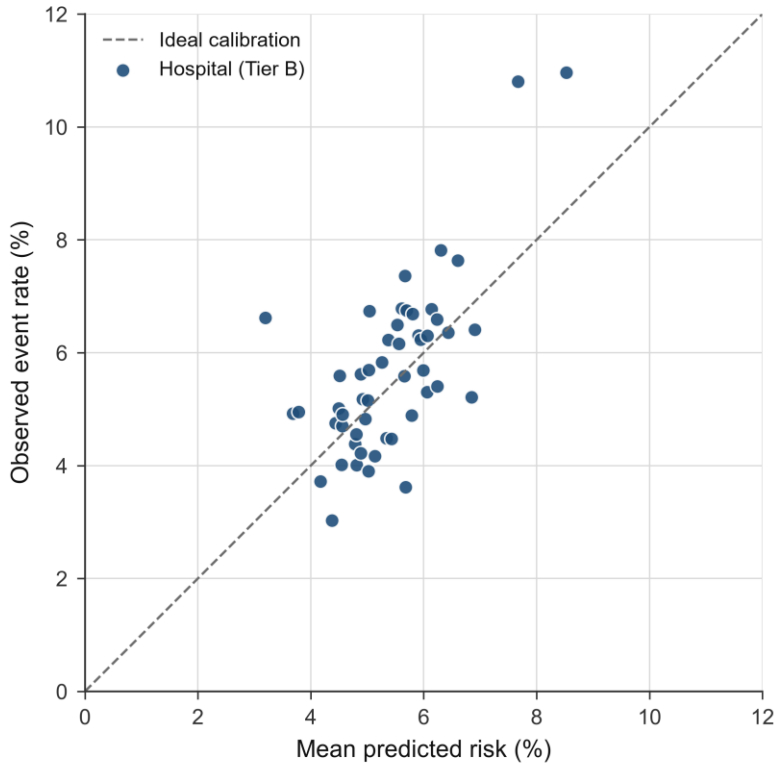


Fig. S2 Hospital-level calibration-in-the-large among 51 hospitals with at least 20 events and 20 non-events. Each point represents an eligible hospital; the dashed diagonal denotes ideal calibration

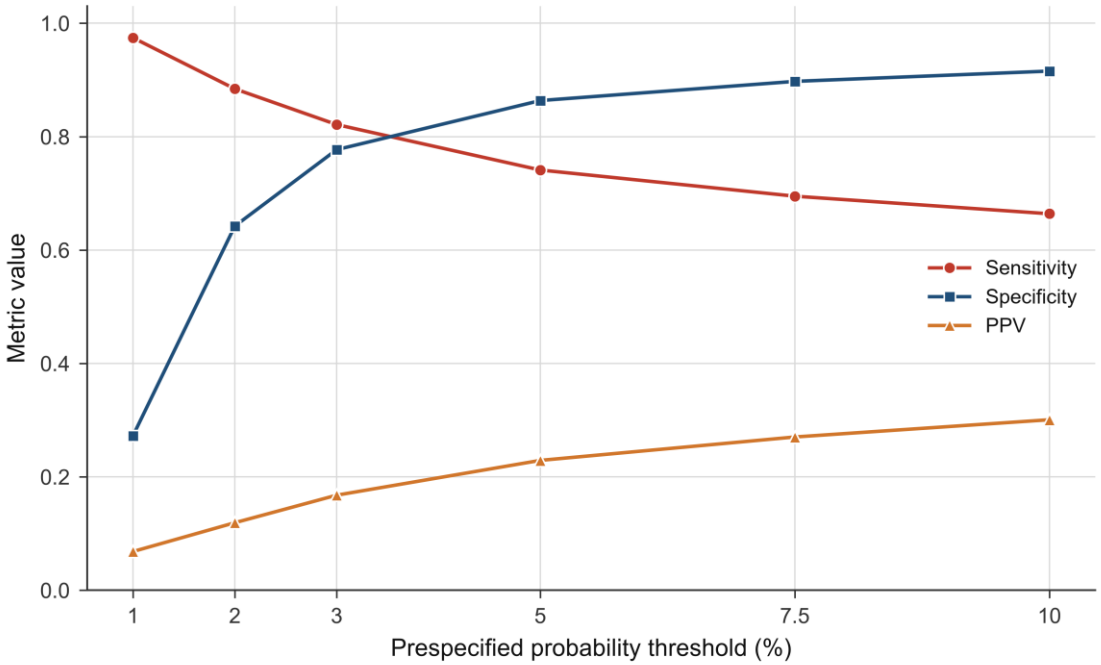


Fig. S3 Operating-point trade-offs for calibrated XGBoost across prespecified probability thresholds. Sensitivity, specificity, and positive predictive value are shown

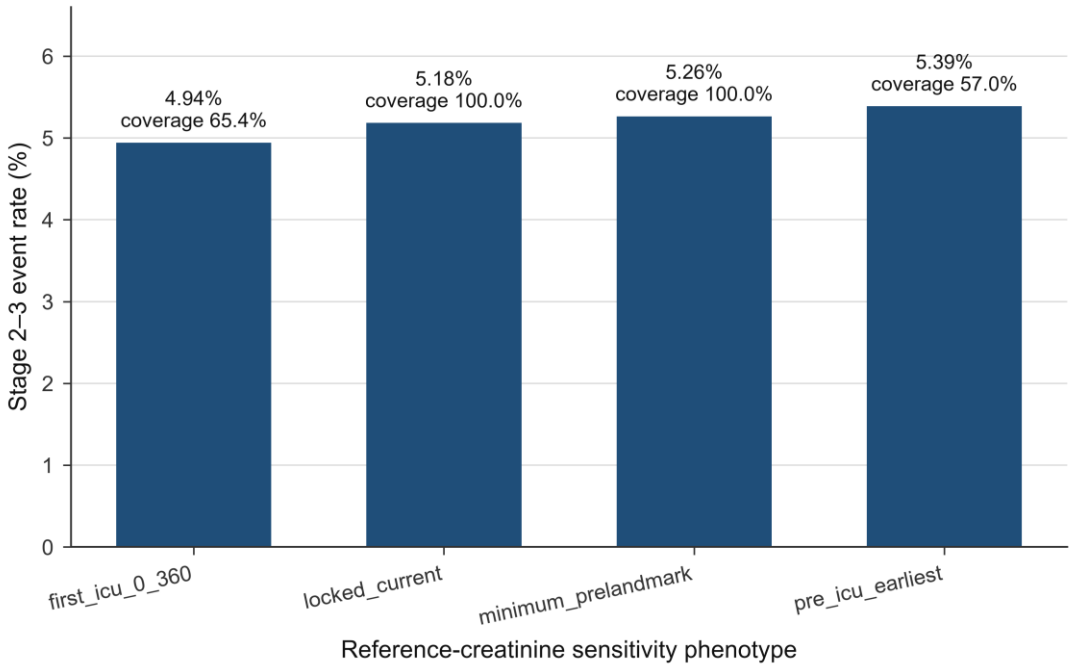


Fig. S4 Stage 2–3 event rates across alternative reference-creatinine definitions. Bars show serum-creatinine-defined event rates; reference-creatinine coverage is annotated

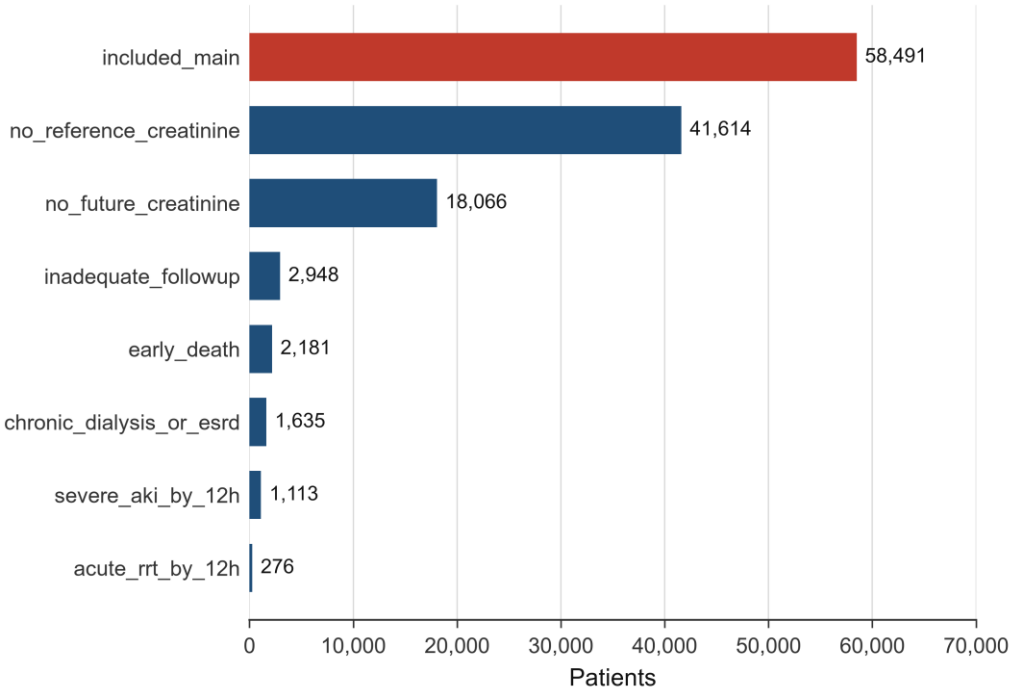


Fig. S5 Mutually exclusive landmark cohort inclusion and exclusion groups. Bars show patient counts within the landmark candidate set; the final primary cohort is shown in red

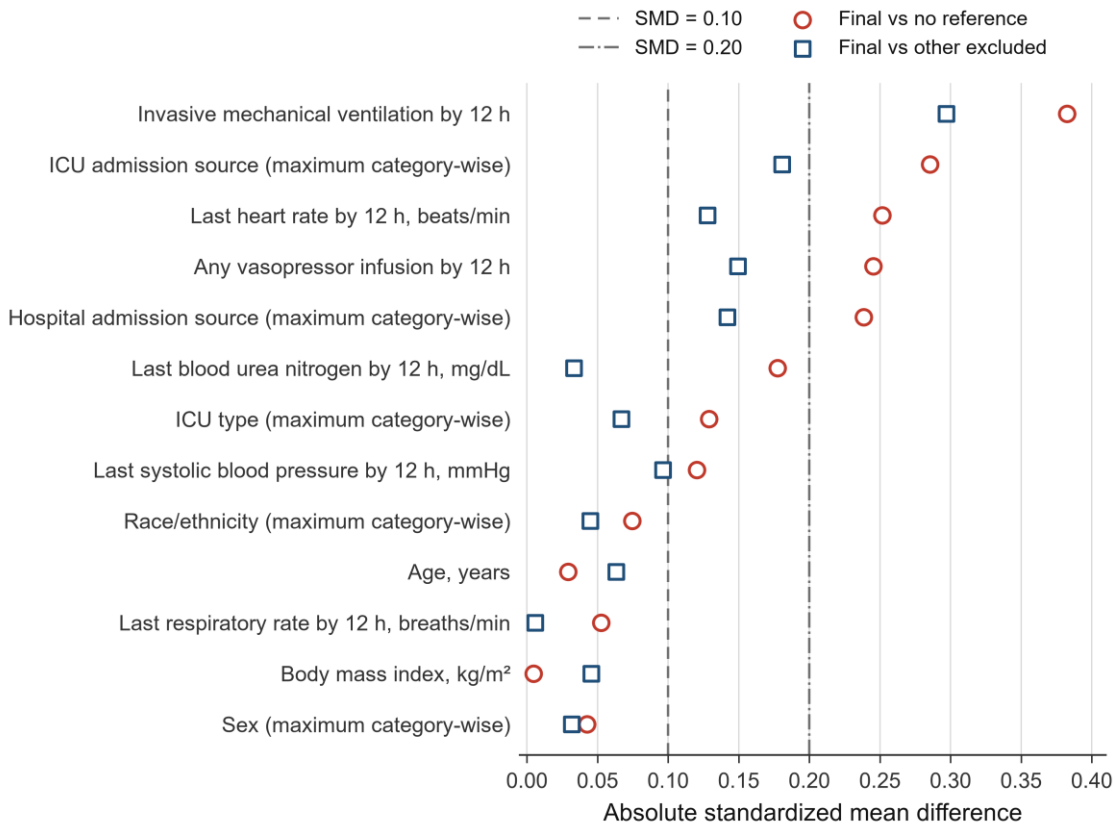


Fig. S6 Covariate shift between the final landmark cohort and excluded candidate populations. Open circles compare the final cohort with patients lacking reference creatinine and open squares compare the final cohort with other excluded patients. Absolute standardized mean differences are descriptive; the 0.10 and 0.20 lines are reference values, not inferential thresholds

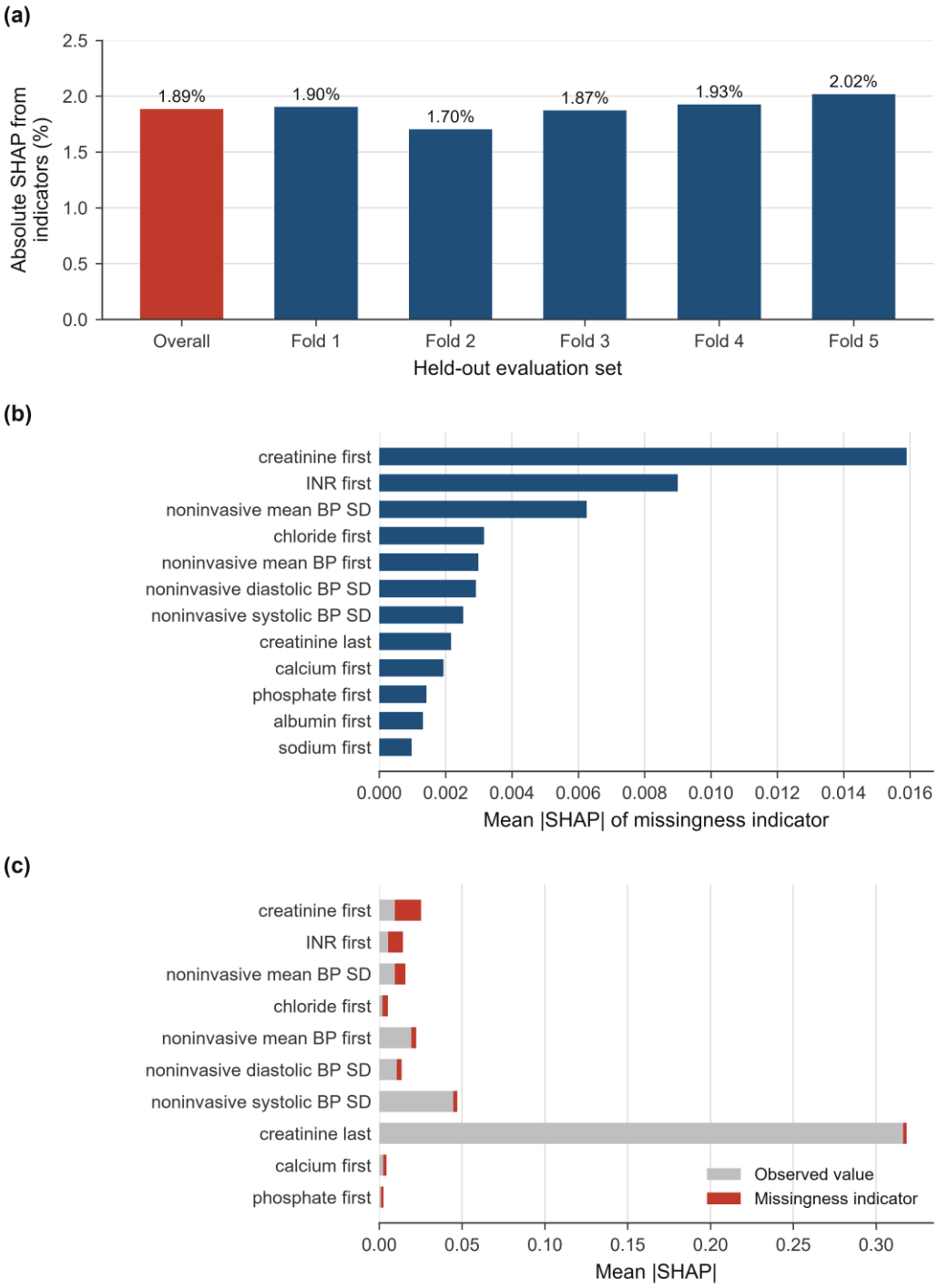


Fig. S7 Explicit numeric missingness-indicator attribution in held-out XGBoost predictions. (a) Overall and outer-fold-specific proportions of total absolute TreeSHAP attribution assigned to explicit missingness indicators. (b) Leading source predictors ranked by mean absolute indicator attribution. (c) Observed-value and missingness-indicator attribution components for the leading source predictors. The aggregate 1.89% is an attribution proportion, not model performance; SHAP attribution is non-causal

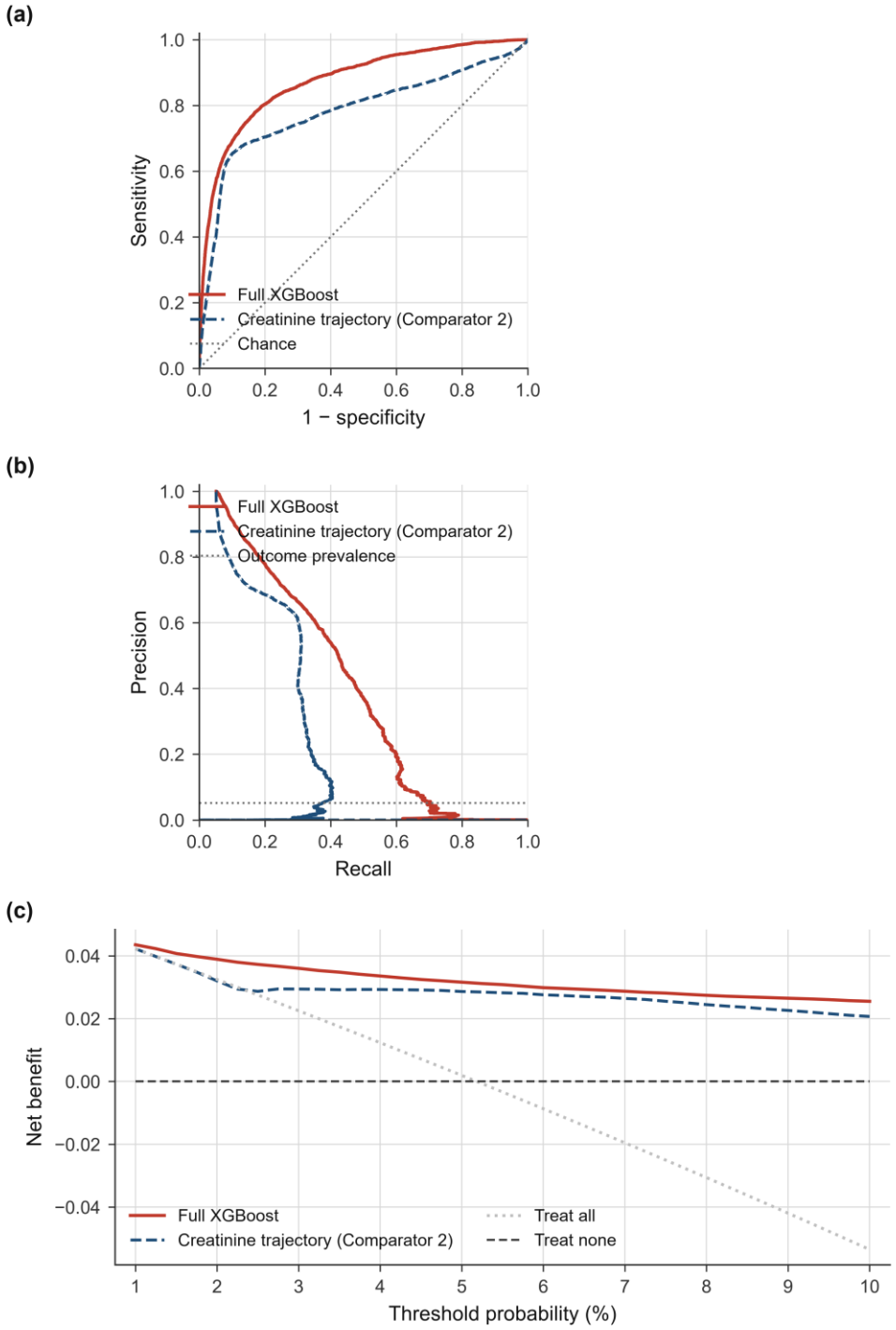


Fig. S8 Incremental predictive value of full XGBoost relative to the prespecified creatinine-trajectory comparator. (a) ROC curves. (b) Precision-recall curves with outcome prevalence shown as the no-skill reference. (c) Decision-curve analysis across 1–10% thresholds. Decision-curve results represent estimated net benefit, not demonstrated outcome benefit

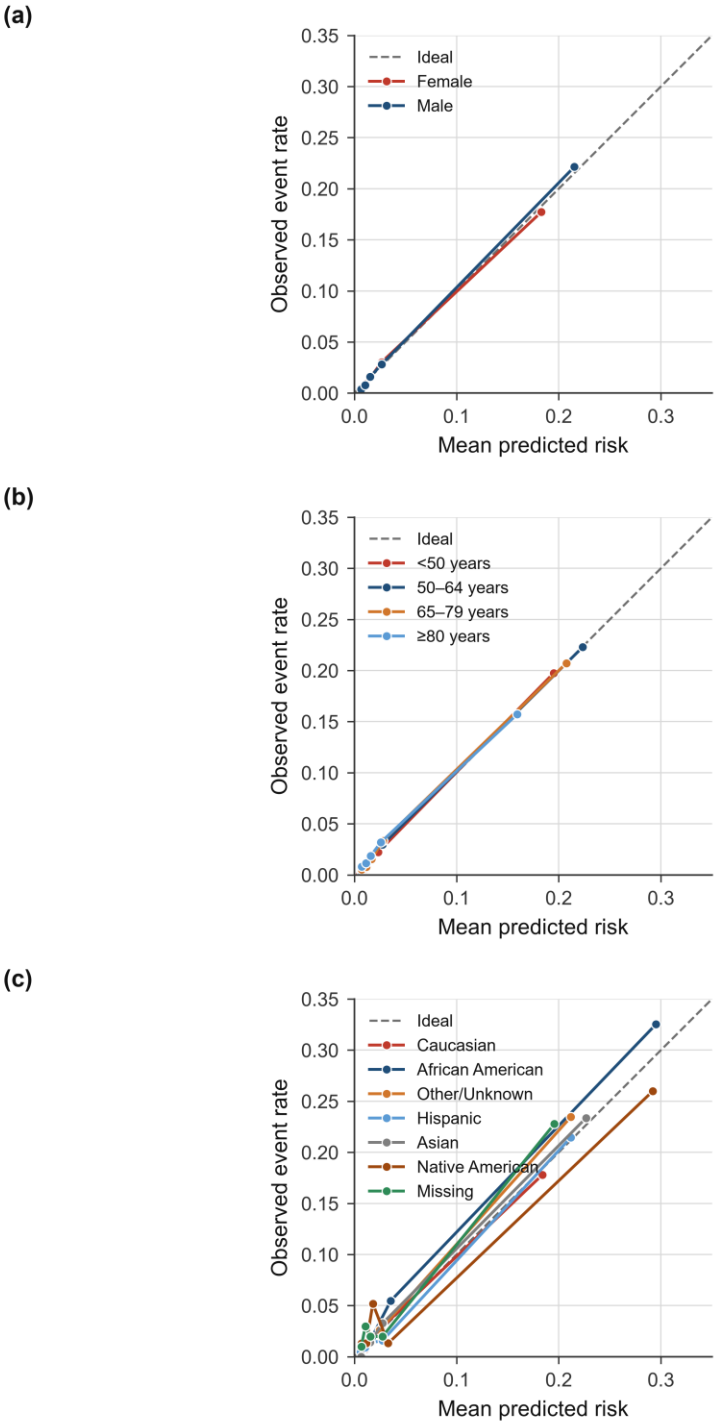


Fig. S9 Exploratory subgroup calibration of original held-out XGBoost predictions without subgroup refitting or recalibration. (a) Sex, (b) age group, and (c) recorded race/ethnicity; each subgroup curve connects calibration quintiles and the dashed diagonal denotes ideal calibration. Race/ethnicity labels reflect recorded database categories and are not biological constructs

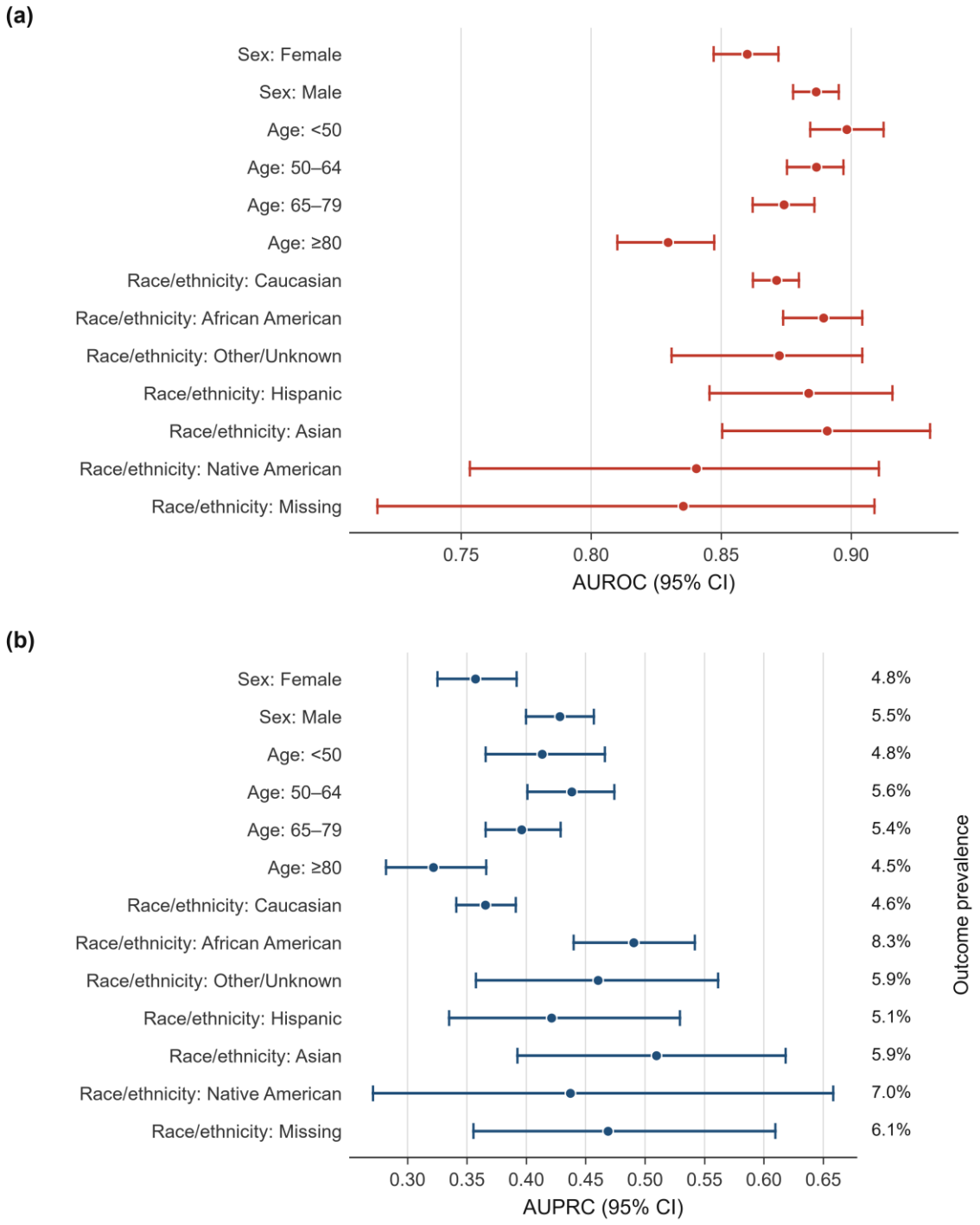


Fig. S10 Exploratory subgroup discrimination and precision-recall performance with prevalence context. (a) AUROC and 95% hospital-cluster bootstrap confidence intervals. (b) AUPRC and 95% confidence intervals; right-side annotations show subgroup outcome prevalence. Hospital-cluster bootstrap uncertainty used 2,000 replicates where estimable; results do not establish fairness or absence of bias

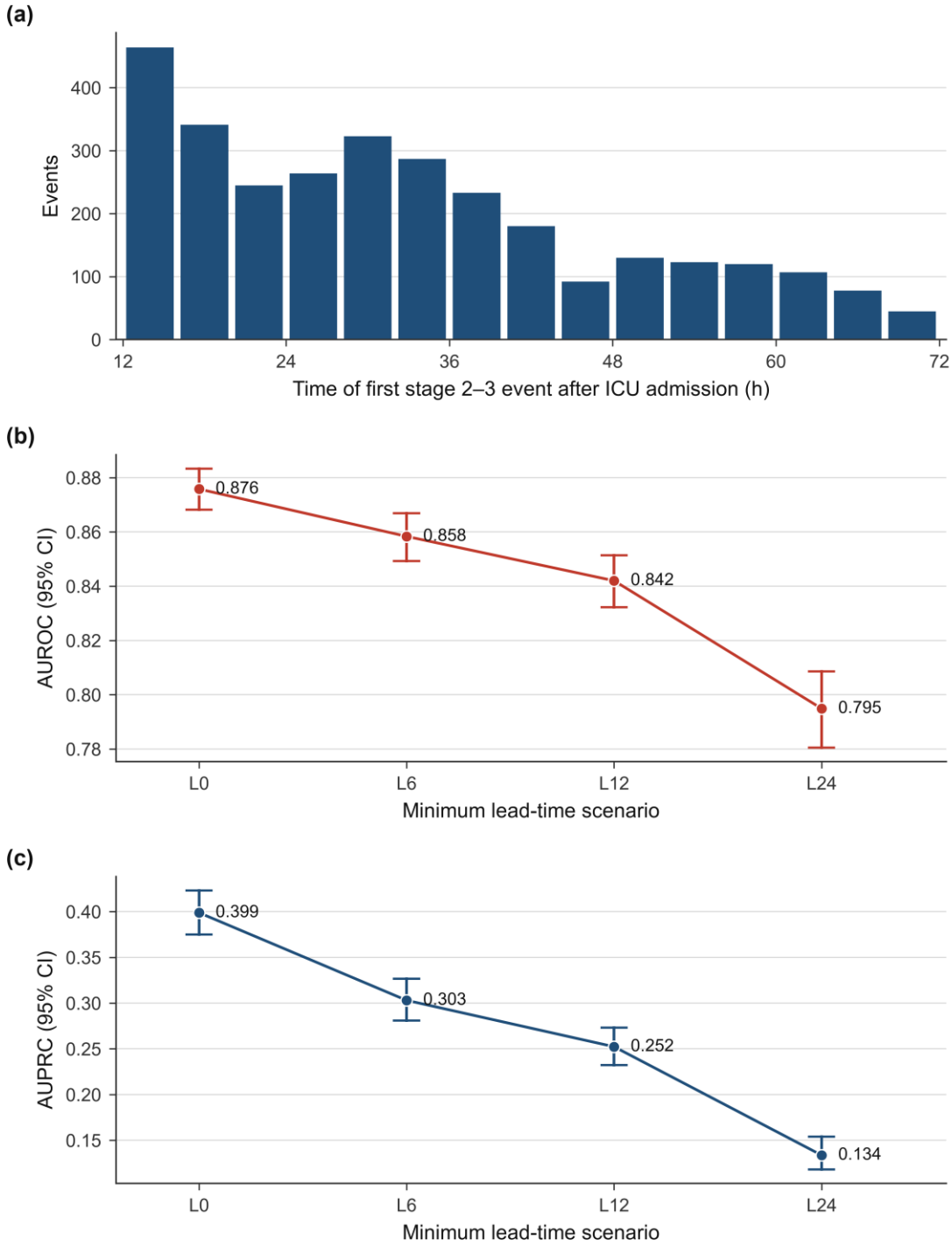


Fig. S11 Actionable lead-time sensitivity using unchanged held-out predictions. (a) Distribution of first qualifying stage 2–3 event times after the 12-h landmark. (b) AUROC and 95% confidence intervals across minimum lead-time scenarios L0, L6, L12, and L24. (c) Corresponding AUPRC and 95% confidence intervals

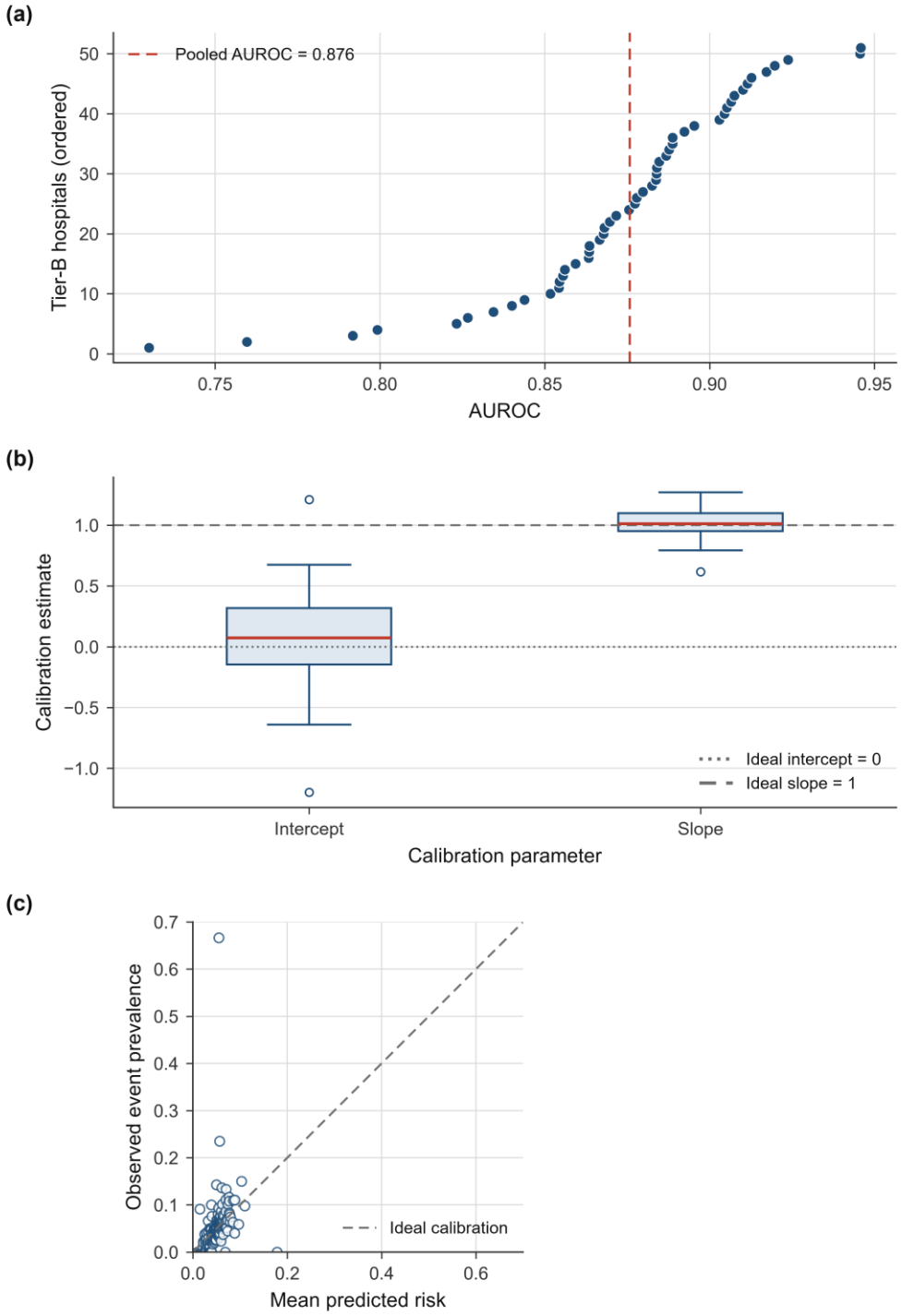


Fig. S12 Hospital-level held-out discrimination and calibration heterogeneity. (a) Ordered AUROC values across 51 Tier-B hospitals with at least 20 events and 20 non-events; the dashed line shows pooled AUROC. (b) Distributions of hospital-specific calibration intercept and slope; reference lines denote ideal intercept = 0 and ideal slope = 1. (c) Observed event prevalence versus mean predicted risk across all 198 held-out hospitals; the dashed diagonal denotes ideal calibration. Raw hospital identifiers are not displayed